STAT 361 Assignment 1 Question 4

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a) Density of Y_i

From Chapter 2, Slide 4 for STAT 361, we know from the assumption that $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, it follows that:

- $E(Y_i|x_i) = \beta_0 + \beta_1 x_i$
- $Var(Y_i) = \sigma^2$
- Y_i follows a normal distribution

Given that Y_i (or equivalently $Y_i|x_i$) follows a normal distribution, then we can correlate that to having $E(Y_i|x_i) = \mu_i$. The density function for a normal distribution $N(\mu_i, \sigma^2)$ as per STAT 268 is:

$$f(x)_{\mu_i} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu_i)^2}{2\sigma^2}\right\}$$

By simple substitution for μ_i and $f(x) = f(y_i|x_i, \beta_0, \beta_1, \sigma^2)$, we have that the density for Y_i is:

$$f(y_i|x_i, \beta_0, \beta_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - \beta_0 + \beta_1 x_i)^2}{2\sigma^2}\right\}$$

$\therefore$ **We have found the density for $f(y_i|x_i, \beta_0, \beta_1, \sigma^2)$ above given that ϵ_i is i.i.d. from a normal distribution.**

(Answer 4a)

b) MLEs of (β_0, β_1)

From STAT 361's Chapter 2, Slide 9, we know that:

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} [Y_i - (\beta_0 + \beta_1 x_i)]^2$$

From the answer to **4a)**, we know that:

$$f(y_i|x_i, \beta_0, \beta_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - \beta_0 + \beta_1 x_i)^2}{2\sigma^2}\right\}$$

and thus

$$\begin{aligned}
L(\beta_0, \beta_1) &= \prod_{i=1}^{n} f(y_i|x_i, \beta_0, \beta_1, \sigma^2) \\
&= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - \beta_0 + \beta_1 x_i)^2}{2\sigma^2}\right\} \\
&= (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\sum_{i=1}^{n}(y_i - \beta_0 + \beta_1 x_i)^2}{2\sigma^2}\right\} \\
&= (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right\}
\end{aligned}$$

In order to produce maximum likelihood estimates, similar to the derivation for Least Squares Estimates in STAT 361's Chapter 2, Slides 5-9, We want to find a $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize $S(\beta_0, \beta_1)$. Minimizing $S(\beta_0, \beta_1)$ will maximize $L(\beta_0, \beta_1)$ as a lower magnitude negative exponent t in $L(\beta_0, \beta_1) = (2\pi\sigma^2)^{-n/2} e^t$ will produce a higher value by elementary exponential rules.

Thus, we may utilize the end result of the derivation for the least squares estimates involving $S(\beta_0, \beta_1)$, letting $\hat{\beta_0}$ and $\hat{\beta_1}$ be estimates of β_0 and β_1, from STAT 361's Chapter 2, Slides 5-9 to find that

$$\hat{\beta_1} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(Y_i - \overline{Y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$
$$\hat{\beta_0} = \overline{Y} - \hat{\beta_1}\overline{x}$$

where $\overline{Y} = \sum_{i=1}^{n} \frac{Y_i}{n}$ and $\overline{x} = \sum_{i=1}^{n} \frac{x_i}{n}$.

$\therefore$ **The maximum likelihood estimates for** (β_0, β_1) **are exactly the least squares estimates as derived in STAT 361's Chapter 2, Slides 5-9** $(\hat{\beta_0}, \hat{\beta_1})$**.**

(Answer 4b)